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Convex Combination Star Shape Prior for Data-driven Image Semantic Segmentation

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Abstract

Multi-center star shape is a prevalent object shape feature, which has proven effective in model-based image segmentation methods. However, the shape field function induced by the multi-center star shape is non-smooth, and directly applying it to the data-driven image segmentation network architecture design may lead to instability in backpropagation. This paper proposes a convex combination star (CCS) shape, possessing multi-center star shape properties, and has the advantage of effectively controlling the shape of the region through a smooth field function. The sufficient condition of the proposed CCS shape can be combined into the image segmentation neural network structure design through the bridge between the variational segmentation model and the activation function of the data-driven method. Taking Segment Anything Model (SAM) and its improved version as backbone networks, we have shown that the segmentation network architecture with CCS shape properties can greatly improve the accuracy of segmentation results.

1. Introduction

Image segmentation is essential in medical and remote sensing fields. Unlike natural images, objects in medical and remote sensing images exhibit distinct geometric shape, so a satisfactory segmentation result should maintain these characteristics. Some task-specific traditional methods, such as graph cuts and active contours [1, 3, 20, 27], improve segmentation accuracy by incorporating shape and other priors [2, 13, 30, 31, 33, 35]. The star-shaped prior, in particular, has shown effective in various studies [14, 24, 36]. Vicente [36] introduced a single-center star-shaped constraint within an interactive graph-cut framework, and later extended by Gulshan *et al*. [14] and Isack *et al*. [18] to multi-center and hedgehog shape models, respectively. However, the interactive graph-cut framework restricts parallel processing, posing efficiency challenges. Additionally, traditional methods typically rely on manually selected low-level features, such as edges, textures, and colors, making it difficult to capture high-level semantic information in complex images. In contrast, deep learning methods [29, 32, 34, 39] can learn more complex and abstract features from large-scale annotated data in an end-to-end manner.

Currently, many studies have integrated shape, boundary, and other priors into neural networks [5, 6, 11, 15, 23], generally falling into two categories: one is adding regularization terms to the loss function [9, 15, 17, 23], while the other is integrating prior information directly into the network architecture via specialized sub-networks [5, 37, 38]. However, regularized-loss-function methods only enforce shape priors during training, with no theoretical guarantee that the network's output retains desired shape properties during prediction. Thus, methods integrating prior information directly into the architecture deserve more attention. For instance, Chen *et al*. [5] refined segmentation predictions by learning directional maps, while works [6, 37] used predicted vertices or contour points to improve segmentation. Other approaches [11, 38] combine boundary information to enhance model performance. However, such methods are often directly implemented by adding network layers based on engineering experience, lacking mathematical interpretability.

Unlike traditional variational models, data-driven methods require backpropagation of gradients during training. It

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is necessary to make modifications to classical variational image segmentation models (especially non-smooth variational models) to incorporate data-driven methods. For instance, variational models with star shape constraints [22], can generate star shapes via a shape field function [18, 22]. Using the smooth field function of the original single-center star shape constraint [36], this constraint can be seamlessly integrated into the segmentation network design in [22]. However, many objects lack a single-center star shape, and imposing such strong priors may lead to undesired star-shaped segmentations. In contrast, multi-center star shapes have a property possessed by the vast majority of objects. Unfortunately, due to the multi-center, the field function that controls the shape is discontinuous and difficult to implement.

To integrate multi-center shape priors into data-driven image segmentation approaches, we develop a novel star shape called the convex combination star (CCS) shape, possessing multi-center star shape properties, and controllable by a smooth flow field. Thus, it can be seamlessly incorporated into variational models and data-driven method. By dual algorithm and unrolling technique, we propose a novel variational network module, termed the CCS module or more specifically the CCS softmax/sigmoid module, as the solution of the dual algorithm takes the form of softmax/sigmoid functions. This allows multi-center star-shaped prior to be directly embedded in segmentation networks through the proposed network module. Different from the existing subnetwork structures designed based on experience, our proposed CCS module fundamentally optimizes a variational problem incorporating the CCS shape constraint. This enables the network to explicitly solve the variational problem through this module during forward propagation, thereby achieving an interpretable fusion of the shape prior field with the segmentation output. Furthermore, we design a novel shape loss function based on CCS shape constraint, which can be directly integrated into the network's loss function to enhance segmentation performance.

To evaluate our methods, we adopt the Segment Anything Model (SAM) [21] and its improved version [28] as backbone networks. Experimental results demonstrate the effectiveness of our CCS prior, shape loss, and CCS module, particularly on medical imaging datasets. We present a segmentation result comparison produced by SAM [21] without and with our CCS module in Fig. 1, showing that the dice accuracy improves by more than 10% with the proposed CCS shape prior incorporating with SAM. Our contributions are summarized as follows:

- We propose a smooth field to constrain the object shape to an convex combination star (CCS), a shape prior possessed by majority of objects. The smoothness of the field theoretically guarantees that after incorporating the CCS shape into the image segmentation network structure design, backpropagation can be executed stably.
- We present a variational model and its dual algorithm based on the constraint of a smooth field for CCS shapes, which allows the multi-center star shape prior to be designed into the image segmentation network structure.
- Through the mathematical connection between the activation function and the variational segmentation model, we propose a CCS softmax/sigmoid network module (Fig. 3), which can be incorporated into any segmentation network with softmax or sigmoid activation.

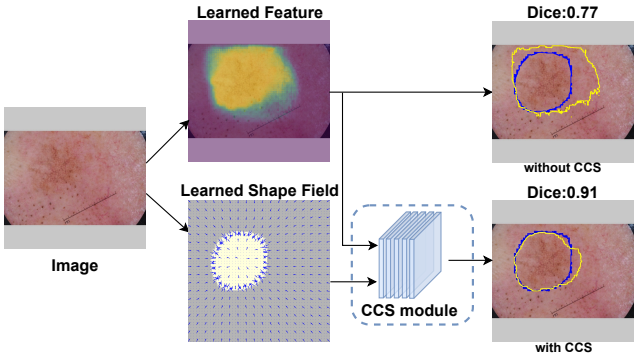


Figure 1. Comparison of segmentation results from SAM without and with the proposed CCS shape knowledge. The blue and yellow lines denote the ground truth and segmentation result contours, respectively. The vector field is learnt by the proposed method.

2. Related work

2.1. Segment anything model

The Segment Anything Model (SAM) [21] is a powerful, general-purpose model for image segmentation, which composed of an image encoder $\mathcal{E}_1$, a prompt encoder $\mathcal{E}_2$ and a mask decoder $\mathcal{D}$. Let $I : \Omega \to \mathbb{R}^d$ be an image over the pixel domain Ω, where $d = 1$ for grayscale and $d = 3$ for color images. The image encoder, a Vision Transformer (ViT) [12], maps I into an image embedding $\tilde{I}$. The mask decoder $\mathcal{D}$ fuses the learnable mask tokens T_m with prompt embeddings $\tilde{P}$ and $\tilde{I}$ to produce segmentation features o, which are subsequently processed by the threshold function $\mathcal{F}$ to generate the final mask $u : \Omega \to \{0, 1\}$. In summary, SAM [21] can be formulated by the following equation:

$$\begin{cases} \tilde{I} = \mathcal{E}_1(I; \Theta_I), \tilde{P} = \mathcal{E}_2(P; \Theta_P), \\ o = \mathcal{D}(\tilde{I}, \tilde{P}; T_m, \Theta_M), \\ u = \mathcal{F}(o). \end{cases} \tag{1}$$

Here, $\Theta = (\Theta_I, \Theta_P, \Theta_M)$ is the parameters of the network. $\mathcal{F}$ is a Heaviside step function, which can be equivalently formulated as the following optimization problem Eq. (2).

$$\mathcal{F}(o) = \arg\min_{u \in \{0,1\}} \{\langle -o, u \rangle\}. \tag{2}$$

While SAM exhibits strong segmentation performance, its prompt embedding mainly provides target object positional information, and studies [41] indicate a tendency to prioritize texture over shape. Recently, Meta released SAM2 [28] for image and video segmentation tasks. When applied to image segmentation, SAM2 adopts a similar architecture to SAM but with a Hiera-based image encoder pre-trained using MAE, enhancing multi-scale feature capture. There are also some variants of SAM designed for specific tasks [4, 7, 8, 37, 40]. However, these modifications lack clear mathematical theoretical foundations. Compared to these methods, we introduce a convex combination star (CCS) shape prior into the model using variational theory.

2.2. Star shape

The concept of star-shaped regions was first introduced by Veksler [36], defining a region as star-shaped with respect to a center $\boldsymbol{c}$ if any ray originating from $\boldsymbol{c}$ intersects the boundary exactly once. To ensure consistency in expression, we write the star shape in the following mathematical definition:

Definition 1 (star shape [36]) *Let $\mathbb{S} \subseteq \mathbb{R}^2$ be a bounded closed region. Given a center $\boldsymbol{c} \in \mathbb{S}$, the region $\mathbb{S}$ is called star shape w.r.t $\boldsymbol{c}$ if the line segment $\{\boldsymbol{y} \mid \boldsymbol{y} = (1-\lambda)\boldsymbol{x} + \lambda\boldsymbol{c}, \lambda \in (0,1)\} \subseteq \mathbb{S}$ for all $\boldsymbol{x} \in \mathbb{S}$.*

In [36], the star-shaped prior was expressed as a discrete energy function which can be optimized using graph cut methods. In [22], the authors extended this star-shaped constraint to a continuous formulation. For a smooth indicator function $u : \Omega \to [0,1]$, they constrained u to the following set $\mathbb{P}_{\boldsymbol{c}}$, enforcing a continuous star-shaped condition:

$$\mathbb{P}_{\boldsymbol{c}} = \{u : \langle \nabla u(\boldsymbol{x}), \boldsymbol{c} - \boldsymbol{x} \rangle \geqslant 0, a.e. \forall \boldsymbol{x} \in \Omega\}. \quad (3)$$

This star-shaped constraint, however, relies on a single-center point, while objects in real images often do not follow a single-center star shape. A natural extension is to use multiple points to define a more general star-shaped region.

Gulshan *et al.* [14] extended the definition of single-center star-shaped to multi-center star-shaped, defined as follows:

Definition 2 (multi-center star shape [14]) *Let $\mathbb{S} \subseteq \mathbb{R}^2$ be a bounded closed region. Given a closed set of star centers $\mathbb{C} \subseteq \mathbb{S}$, the region $\mathbb{S}$ is called multi-center star shape w.r.t $\mathbb{C}$ if $\forall \boldsymbol{x} \in \mathbb{S}, \exists \boldsymbol{c}(\boldsymbol{x}) \in \arg\min_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2$, such that the line segment $\{\boldsymbol{y} \mid \boldsymbol{y} = (1-\lambda)\boldsymbol{x} + \lambda\boldsymbol{c}(\boldsymbol{x}), \lambda \in (0,1)\} \subseteq \mathbb{S}$.*

The point set $\mathbb{C}$ can be finite or infinite. Gulshan *et al.* [14] incorporated this multi-center star-shaped constraint into interactive segmentation via graph cuts. However, the center point $\boldsymbol{c}(\boldsymbol{x})$ in Definition 2 may not be unique, potentially resulting the corresponding multi-center shape field being non-smooth when using continuous field conditions to control the shape. This non-smooth field further leads to issues in the backpropagation. Therefore, in the next section, we introduce a new star shape, allowing the multi-center star shape prior constraint to be integrated into neural networks.

3. Method

3.1. Convex combination star shape

In order to obtain a smooth field constraint condition for a multi-center star shape, we first give the definition of the proposed convex combination star shape. In most cases, the point set $\mathbb{C}$ is discrete. Therefore, we employ summation in our formulation. However, this approach can be naturally extended to integration for more general scenarios.

Definition 3 (CCS shape) *Let $\mathbb{S} \subseteq \mathbb{R}^2$ be a bounded closed region. Given a closed set of multi-point $\mathbb{C} \subseteq \mathbb{S}$, let the convex combination center set of $\mathbb{C}$ be*

$$\mathbb{C}_\varepsilon = \left\{ \boldsymbol{c}_\varepsilon(\boldsymbol{x}), \forall \boldsymbol{x} \in \mathbb{S} : \boldsymbol{c}_\varepsilon(\boldsymbol{x}) = \sum_{\boldsymbol{z} \in \mathbb{C}} w_\varepsilon^*(\boldsymbol{z}) \boldsymbol{z} \right\}. \quad (4)$$

where

$$w_\varepsilon^*(\boldsymbol{z}) := \frac{e^{\frac{-\|\boldsymbol{x}-\boldsymbol{z}\|^2}{\varepsilon}}}{\sum_{\boldsymbol{z}' \in \mathbb{C}} e^{\frac{-\|\boldsymbol{x}-\boldsymbol{z}'\|^2}{\varepsilon}}}. \quad (5)$$

The region $\mathbb{S}$ is called convex combination star (CCS) shape w.r.t $\mathbb{C}$ if the line segment $\{\boldsymbol{y} \mid \boldsymbol{y} = (1-\lambda)\boldsymbol{x} + \lambda\boldsymbol{c}_\varepsilon(\boldsymbol{x}), \lambda \in (0,1)\} \subseteq \mathbb{S}$ holds for all $\boldsymbol{x} \in \mathbb{S}$.

Next, we provide the main mathematical derivation framework and the motivation of CCS construction. We first present the following duality and smooth approximation propositions for the $\min$ function:

Proposition 1 (duality of $\min$) *For the concave function $\min$, we have*

$$\min_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2 = \min_{w \in \mathbb{W}} \sum_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2 w(\boldsymbol{z}), \quad (6)$$

where $\mathbb{W} = \{w(\boldsymbol{z}) \in [0,1] : \sum_{\boldsymbol{z} \in \mathbb{C}} w(\boldsymbol{z}) = 1\}$. Furthermore, if there is a unique minimizer of $\min_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2$, denote $w^* = \arg\min_{w \in \mathbb{W}} \sum_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2 w(\boldsymbol{z})$, we have $\boldsymbol{c}(\boldsymbol{x}) = \sum_{\boldsymbol{z} \in \mathbb{C}} w^*(\boldsymbol{z}) \boldsymbol{z}$.

We use the following smooth technique for $\min$ function appeared in Definition 2.

Proposition 2 (smooth $\min$) *The ε-smooth $\min$ is defined as:*

$$\min_{\boldsymbol{z} \in \mathbb{C}}{}_\varepsilon(\|\boldsymbol{x} - \boldsymbol{z}\|^2) := -\varepsilon \ln \left(\sum_{\boldsymbol{z} \in \mathbb{C}} e^{-\frac{\|\boldsymbol{x}-\boldsymbol{z}\|^2}{\varepsilon}} \right). \quad (7)$$

Then we have $\lim_{\varepsilon \to 0^+} \min_{\boldsymbol{z} \in \mathbb{C}}{}_\varepsilon(\|\boldsymbol{x} - \boldsymbol{z}\|^2) = \min_{\boldsymbol{z} \in \mathbb{C}} \|\boldsymbol{x} - \boldsymbol{z}\|^2$.

Proposition 3 (duality of $\min_\varepsilon$) *$\min_\varepsilon$ is concave and its duality*

$$\min_{\boldsymbol{z}\in\mathbb{C}}{}_\varepsilon(\|\boldsymbol{x}-\boldsymbol{z}\|^2) = \min_{w\in\mathbb{W}} \left\{ \begin{array}{l} \displaystyle\sum_{\boldsymbol{z}\in\mathbb{C}} \|\boldsymbol{x}-\boldsymbol{z}\|^2 w(\boldsymbol{z}) \\ \quad + \varepsilon \displaystyle\sum_{\boldsymbol{z}\in\mathbb{C}} w(\boldsymbol{z}) \ln w(\boldsymbol{z}) \end{array} \right\}. \tag{8}$$

Here $\mathbb{W}$ is the same as Proposition 1. Furthermore, let w_ε^* represent the minimizer of the right optimization problem, then

$$w_\varepsilon^*(\boldsymbol{z}) := \frac{e^{\frac{-\|\boldsymbol{x}-\boldsymbol{z}\|^2}{\varepsilon}}}{\displaystyle\sum_{\boldsymbol{z}'\in\mathbb{C}} e^{\frac{-\|\boldsymbol{x}-\boldsymbol{z}'\|^2}{\varepsilon}}}. \tag{9}$$

Proofs of these propositions are provided in Appendix A.

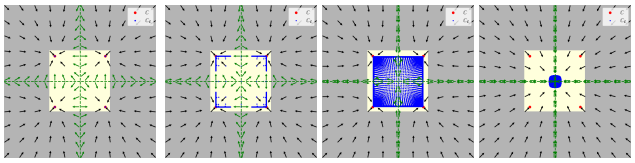
In Proposition 3, we cannot obtain the analytical expression of the minimizer $\boldsymbol{c}_\varepsilon(\boldsymbol{x})$ in terms of the dual variables w_ε^*. However, compared to Proposition 1, when ε is sufficiently small, we still use $\boldsymbol{c}_\varepsilon(\boldsymbol{x}) = \sum_{\boldsymbol{z}\in\mathbb{C}} w_\varepsilon^*(\boldsymbol{z})\boldsymbol{z}$ as the minimizer, yielding $\boldsymbol{c}_\varepsilon(\boldsymbol{x})$ as defined in Definition 3.

These propositions show that CCS is essentially obtained by smoothing the $\min$ function. Compared to the non-strict concavity and non-smoothness of the $\min$ function, $\min_\varepsilon$ is strictly concave and smooth, which ensures the uniqueness of $\boldsymbol{c}_\varepsilon(\boldsymbol{x})$. However, in Definition 2, $\boldsymbol{c}(\boldsymbol{x})$ is not necessarily unique. This property is very important for us to define the shape field using $\boldsymbol{c}_\varepsilon(\boldsymbol{x})$ later.

In the above Definition 3, it is noted that $\mathbb{C}_\varepsilon$ is the convex combination of points in $\mathbb{C}$, so $\mathbb{C}_\varepsilon$ is a subset of the convex hull of $\mathbb{C}$. The parameter ε controls the size of the set $\mathbb{C}_\varepsilon$. When ε approaches 0, the Definition 3 is equivalent to the multi-point centered star shape (Definition 2). When ε approaches positive infinity, the Definition 3 is equivalent to the star shape (Definition 1) centered at the centroid of $\mathbb{C}$. Therefore, the CCS shape we defined is a special star shape between single-centered and multi-centered star shapes. It has the advantages of both types of star shapes. These facts can be summarized as the following mathematical proposition, which can be directly obtained by Proposition 1 and Proposition 3.

Proposition 4 (the essence of CCS) *Let $\boldsymbol{c}_\varepsilon(\boldsymbol{x})$ be defined in Definition 3, $\boldsymbol{c}(\boldsymbol{x})$ be defined in Definition 2 and is unique, then $\lim_{\varepsilon\to 0^+} \boldsymbol{c}_\varepsilon(\boldsymbol{x}) = \boldsymbol{c}(\boldsymbol{x})$, $\lim_{\varepsilon\to+\infty} \boldsymbol{c}_\varepsilon(\boldsymbol{x}) = \hat{\boldsymbol{c}}$, where $\hat{\boldsymbol{c}}$ is the centroid of $\mathbb{C}$.*

To visualize this property of CCS, we show the variation of the center point set $\mathbb{C}_\varepsilon$ with different ε values in Fig. 2. To emphasize the variations in the vector field $\boldsymbol{c}(\boldsymbol{x}) - \boldsymbol{x}$, we highlight the small square region in the center. As shown in Fig. 2, we can observe that the vector field corresponding to the multi-center star shape is discontinuous along the perpendicular bisector, while $\boldsymbol{c}_\varepsilon(\boldsymbol{x}) - \boldsymbol{x}$ is smooth.



(a) $\varepsilon \to 0^+$ (b) $\varepsilon = 1$ (c) $\varepsilon = 10$ (d) $\varepsilon = 100$

Figure 2. Illustration of the vector field $\boldsymbol{c}_\varepsilon(\boldsymbol{x}) - \boldsymbol{x}$ and center point set $\mathbb{C}_\varepsilon$ with different ε.

3.2. Convex combination star shape constraint

From the definition of the CCS shape, it can be seen that most objects and segmentation tasks conform to this shape. We detail this property as Theorem 1.

Theorem 1 (universal shape prior) *Let $\mathbb{S} \subseteq \mathbb{R}^2$ be a bounded closed region, and let $\partial\mathbb{S}$ denote its boundary set. Then, $\mathbb{S}$ is a CCS shape region with respect to $\partial\mathbb{S}$ when $\varepsilon \to 0^+$.*

Theorem 1 states that all objects are star-shaped with multiple centers at their boundaries, and when ε is sufficiently small, they are also CCS shapes.

Directly applying the definition to enforce the CCS shape constraint requires calculating the line segment from each foreground pixel to its corresponding center point. Inspired by [22], we present in Theorem 2 a sufficient condition for CCS shapes, which makes the constraint more computationally efficient to handle.

Theorem 2 (sufficient condition) *Let $u \in C^1(\Omega)$, and assume superlevel set $u^{-1}[\gamma, +\infty)$ is closed with a simple, closed boundary curve. Further, assume the closed set $\mathbb{C} \subseteq u^{-1}[\gamma, +\infty)$. Define the set $\mathbb{P}$ as :*

$$\mathbb{P} = \{u : \langle \nabla u(\boldsymbol{x}), \boldsymbol{s}(\boldsymbol{x})\rangle \geqslant 0, a.e. \forall \boldsymbol{x} \in \Omega\}. \tag{10}$$

where $\boldsymbol{s}(\boldsymbol{x}) = \boldsymbol{c}_\varepsilon(\boldsymbol{x}) - \boldsymbol{x}$. If $u \in \mathbb{P}$, then any superlevel set $u^{-1}[\gamma, +\infty)$ is a CCS shape region with respect to set $\mathbb{C}$ when $\varepsilon \to 0^+$.

Proofs of the theorems could be found in Appendix B.

3.3. Convex combination star shape prior with variational formula

With the sufficient condition from Theorem 2, we can construct an image segmentation model incorporating the CCS prior as follows:

$$\min_{u\in[0,1]\cap\mathbb{P}} \{\langle -o, u\rangle + \mathcal{H}(u)\}. \tag{11}$$

Here o is the segmentation feature function and the set $\mathbb{P}$ is given by Eq. (10). $\mathcal{H}(u) = \xi\left(\langle u, \ln u\rangle + \langle 1-u, \ln(1-u)\rangle\right)$ is the entropy term to ensure smoothness of segmentation function u. In fact, the solution to this optimization problem without the restriction $u \in \mathbb{P}$ is exactly the sigmoid activation function in neural networks. This explains these activation functions with classification functions from the perspective of variational theory. For simplicity of notation, we refer to this model as CCS model.

By introducing a dual variable $q : \Omega \to \mathbb{R}$, the Eq. (11) can be represented by the following dual problem:

$$\min_{u\in[0,1]} \max_{q\geqslant 0} \left\{\langle -o, u\rangle + \mathcal{H}(u) - \langle q, \boldsymbol{s}\cdot\nabla u\rangle\right\}. \tag{12}$$

We update the dual variable q using projected gradient descent. Once q^{t+1} is obtained, we then minimize the subproblem with respect to u to get u^{t+1}. The iterative process is given by:

$$\begin{cases} q^{t+1} = ReLU\left(q^t - \tau_q \boldsymbol{s}\cdot\nabla u^t\right). \\ u^{t+1} = \underset{u\in[0,1]}{\arg\min}\left\{\langle -o + div(q^{t+1}\boldsymbol{s}), u\rangle + \mathcal{H}(u)\right\}. \\ \qquad = sigmoid\left(\dfrac{o - div(q^{t+1}\boldsymbol{s})}{\xi}\right). \end{cases} \tag{13}$$

Here, div refers to the divergence operator, τ_q represents the step size. More details for calculating u^{t+1} can be found in Appendix C. Since we need to compute the divergence of the vector field $\boldsymbol{s}$, the smoothing we introduced earlier is essential.

The solution steps for the original problem Eq. (11) are summarized in Algorithm 1.

Algorithm 1

Input: Feature function o, and the set $\mathbb{C}$.
Output: Soft segmentation function u.

1: Initialize $u^0 = sigmoid(o), q^0 = 0$, and calculate the vector field $\boldsymbol{s}$.
2: **for** $t = 1, 2, \ldots, T$ **do**
3: Update the dual variable q^{t+1} by the first equation of (13).
4: Update u^{t+1} by the second equation of (13).
5: **if** it is converged **then**
6: Break the loop.
7: **end if**
8: **end for**
9: **return** Final solution u.

3.4. Convex combination star shape prior for deep learning

In this section, we use the SAM [21] model as an example to introduce two methods for incorporating our proposed CCS prior: (1) adding a shape loss term to the loss function, and (2) integrating the shape prior through a variational approach. Both methods are applicable to any segmentation network.

3.4.1. Convex combination star shape loss

In image segmentation networks, the cross-entropy loss $\mathcal{L}_{CE}$ is widely adopted. However, it is primarily computed on a pixel-wise basis and fails to leverage more global information. To address this, we propose the following CCS loss $\mathcal{L}_S$ based on the proposed constraint:

$$\mathcal{L}_S(u, \boldsymbol{s}_g) = \sum_{\boldsymbol{x}\in\Omega} ReLU\left(\frac{-\nabla u(\boldsymbol{x})\cdot \boldsymbol{s}_g(\boldsymbol{x})}{\|\nabla u(\boldsymbol{x})\|\|\boldsymbol{s}_g(\boldsymbol{x})\|}\right). \tag{14}$$

where $\boldsymbol{s}_g$ denotes the CCS shape field derived from the ground truth and its computation is elaborated in the Sec. 4.2. As $\mathcal{L}_S(u) \to 0$, u lies in the set $\mathbb{P}$ defined in Eq. (10).

We use Eq. (15) as the loss function and validate its effectiveness on several datasets.

$$\mathcal{L}_{seg} = \alpha_1 \mathcal{L}_{CE} + \alpha_2 \mathcal{L}_S. \tag{15}$$

3.4.2. Convex combination star shape sigmoid (CCS-Sigmoid) module

Different from the existing sub-network structures designed based on experience, we unroll Algorithm 1 into a network module, termed the CCS-Sigmoid module, where each layer operation corresponds to a single iteration.

Given the feature o and vector field $\boldsymbol{s}$, Algorithm 1 produces a segmentation result that preserves the convex combination star shape. Since $\boldsymbol{s}$ is computed from the ground truth, we introduce an learnable token T_f in the decoder to predict the CCS shape field $\boldsymbol{s}$.

Notably, the output u from Algorithm 1 is sigmoid-shaped. Since the iterative process consists of simple mathematical operations, unrolling this algorithm into a network module ensures stable backpropagation. Given that our prior incorporates spatial information, we omit the prompt encoder of SAM. The complete network is illustrated in Fig. 3, with the computation process formalized by the following equation:

$$\begin{cases} \tilde{I} = \mathcal{E}_1(I; \Theta_I), \\ o, \boldsymbol{s} = \mathcal{D}(\tilde{I}; T_m, T_f, \Theta_M), \\ u = \underset{u\in[0,1]\cap\mathbb{P}}{\arg\min}\left\{\langle -o, u\rangle + \mathcal{H}(u)\right\}. \end{cases} \tag{16}$$

Compared to the original decoder Eq. (1), the new model integrates the learned shape field information with segmentation features o through variational model-based (the u subproblem) network layers.

4. Numerical experiments

To validate the effectiveness of the CCS constraint, we compare the following methods: fine-tuning the SAM [21]/

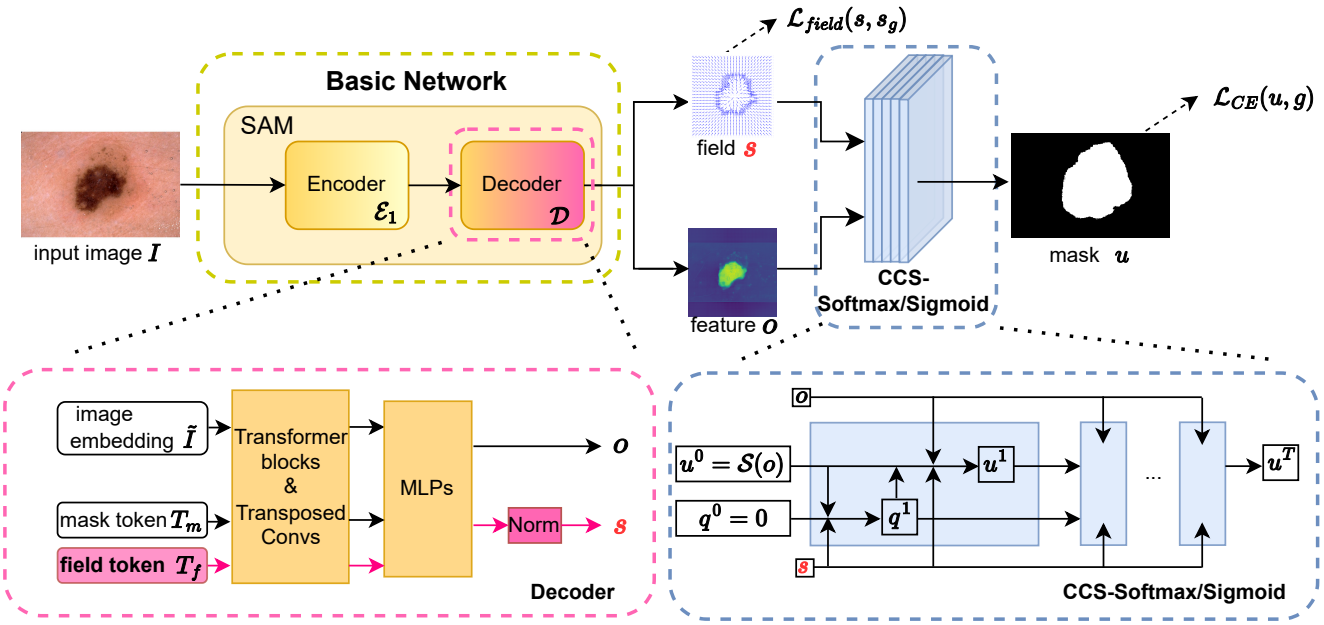


Figure 3. Architecture of the CCS softmax/sigmoid network.

SAM2 [28] with and without CCS loss Eq. (15) and CCS module Fig. 3, respectively.

4.1. Training protocol and experimental setup

While SAM [21] provides three image encoder sizes, we select ViT-B for its balanced performance and computational efficiency. During training, we freeze the image encoder weights, fine-tune them using LoRA [16], and fully train the mask decoder. For SAM2 [28], we selecte Hiera-Base as the decoder and update all parameters used for image segmentation in SAM2.

In CCS module, the CCS-Sigmoid layer count T is set to 10, while the parameters ξ and τ_q are both set to 1. Details can be found in Appendix E. Gradients and divergences are computed via convolutional layers with fixed kernels. As shown in Fig. 3, the loss function of this network consists of two components. The first term optimizes the network's prediction of the CCS field $\boldsymbol{s}$. Given the image region Ω and ground truth field $\boldsymbol{s}_g$, we measure this loss using cosine similarity:

$$\mathcal{L}_{field}(\boldsymbol{s}, \boldsymbol{s}_g) = 1 - \sum_{\boldsymbol{x}\in\Omega} \frac{\langle \boldsymbol{s}(\boldsymbol{x}), \boldsymbol{s}_g(\boldsymbol{x})\rangle}{|\Omega|}. \tag{17}$$

The second term of the loss measures the discrepancy between the segmentation output u and the ground truth g using cross-entropy loss $\mathcal{L}_{CE}$. Overall, the loss function of our network integrated CCS module is expressed as follows:

$$\mathcal{L} = \beta_1 \mathcal{L}_{field} + \beta_2 \mathcal{L}_{CE}. \tag{18}$$

4.2. Datasets and preprocessing

We evaluate our method and model on the following three medical imaging datasets and one remote sensing dataset.

ISIC 2018 dataset [10]: This dataset contains 2,594 high-resolution skin lesion images with segmentation annotations. We use 1,868 images for training, 467 for validation, and the remaining 259 for testing. All images are padded to a square format and resized to 512×512.

Refuge dataset [25]: This dataset consists of 1,200 images, evenly divided into training, validation, and test sets for optic cup and optic disc segmentation. Here, we focus only on optic cup segmentation, with all images cropped to 512×512 around the ROI region.

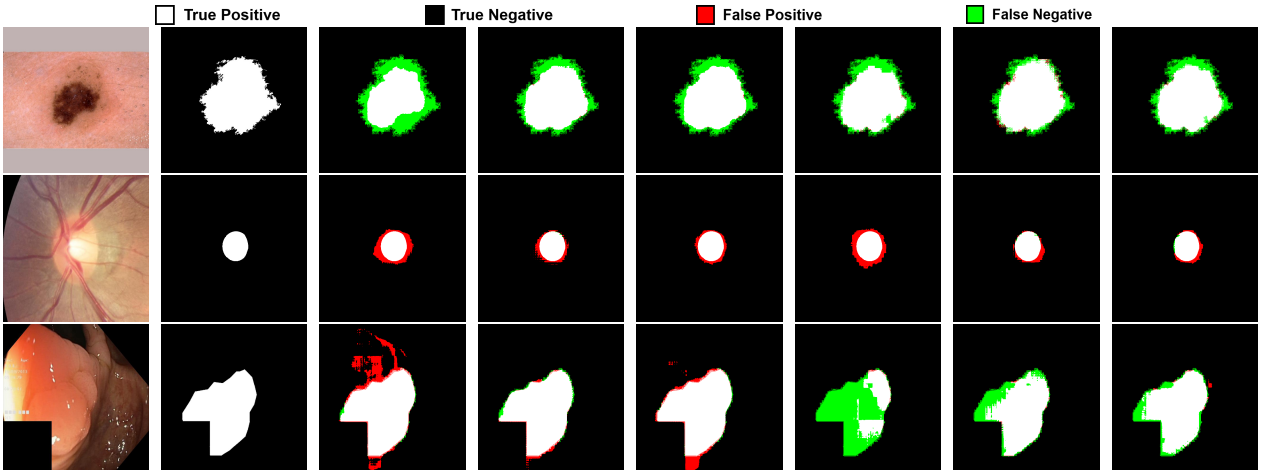
Kvasir dataset [26]: The Kvasir dataset includes 1,000 endoscopy images with polyp segmentation annotations. We divide it into 800 images for training, 100 for validation, and 100 for testing. All images are resized to 512×512.

WHU Building dataset [19]: This dataset contains 8,189 images at 512×512 resolution, captured from Christchurch, New Zealand. It is split into 4,736 images for training, 1,036 for validation, and 2,416 for testing.

The vector field $\boldsymbol{s}_g$ is computed from the set of center points $\mathbb{C}$ in the ground truth, and we propose two straightforward methods to obtain $\mathbb{C}$. For relatively simple-shaped medical images, we uniformly sample 30 points along the boundary. For remote sensing images with more complex shapes, we employ morphological methods to extract points from the convex and concave regions of the image.

Table 1. Results on medical image datasets. The best result is **bold**

backbone	model	ISIC			Refuge			Kvasir		
		DICE↑	*IOU*↑	*ACC*↑	*DICE*↑	*IOU*↑	*ACC*↑	*DICE*↑	*IOU*↑	*ACC*↑
SAM	Finetuned[21]	88.81	81.62	97.40	77.73	64.74	97.93	84.82	77.58	95.62
	CCS loss	89.43	82.16	97.44	79.39	66.87	98.13	85.30	77.99	95.81
	CCS module	**89.67**	**82.45**	**97.52**	**80.82**	**68.96**	**98.31**	**86.91**	**80.51**	**96.15**
SAM2	Finetuned[28]	87.33	79.37	96.95	79.05	66.30	98.13	80.24	70.71	94.31
	CCS loss	**88.32**	80.58	**97.28**	**81.63**	**70.07**	**98.49**	81.27	71.98	**94.87**
	CCS module	88.30	**80.59**	97.08	81.15	69.39	98.38	**82.05**	**73.47**	94.77



(a) Image (b) Ground Truth (c) SAM $_{\text{Finetuned}}$ (d) SAM $_{\text{CCSloss}}$ (e) SAM $_{\text{CCSmodule}}$ (f) SAM2 $_{\text{Finetuned}}$ (g) SAM2 $_{\text{CCSloss}}$ (h) SAM2 $_{\text{CCSmodule}}$

Figure 4. Visual results of medical datasets.

4.3. Results

4.3.1. Inference efficiency

We evaluate the inference efficiency in terms of frames per second (FPS) and floating point operations per second (FLOPs) of SAM2. With the integration of the CCS module, the FPS slightly decreases from 16.55 to 14.42, while the FLOPs remain nearly unchanged (651.34G → 651.38G), demonstrating that the CCS module introduces negligible overhead to inference speed and computational cost.

4.3.2. Medical image segmentation

In Tab. 1, we report the experimental results across three medical image datasets, showing that both approaches enhance the segmentation performance. Fig. 4 visualizes examples, highlighting significant improvements for images with ambiguous boundaries and irregular shapes.

4.3.3. Remote sensing image segmentation

Tab. 2 presents the segmentation results of models on the WHU Building dataset. As shown in Fig. 5, in the presence of tree canopy obstruction or strong light reflection, the building segmentation results are prone to partial omissions, which can disrupt the connectivity of the structures. In contrast, the proposed CCS method can alleviate this issue to some extent, thereby better preserving the shape of the buildings.

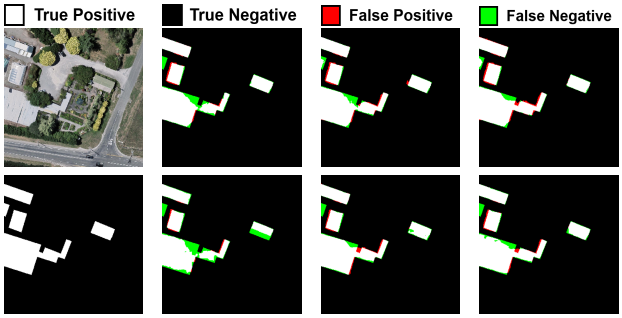
Table 2. Result on the WHU dataset. The best result is **bold**.

backbone	model	*DICE*↑	*IOU*↑	*ACC*↑
SAM	Finetuned[21]	91.60	86.31	**98.42**
	CCS loss	92.08	86.81	98.39
	CCS module	**92.27**	**87.04**	98.40
SAM2	Finetuned[28]	92.33	87.38	98.45
	CCS loss	93.26	**88.58**	98.59
	CCS module	**93.32**	88.56	**98.59**

4.4. Ablation

To explore the optimal integration of the CCS module with the model, we conduct ablation experiments in this section. For more ablation studies, please refer to the Appendix F.

Ablation Study on the Number of CCS Layers: To



(a) Image& GT (b) Finetuned (c) CCS loss (d) CCS module

Figure 5. Visual result of WHU dataset. The first and second rows of (a) show the image and its ground truth segmentation, respectively. The first and second rows of (b), (c), and (d) show the segmentation results of three variants of SAM and SAM2, respectively.

Table 3. Ablation Study on the Number of CCS Layers

	ISIC		
	DICE↑	*IOU*↑	*ACC*↑
$T = 5$	89.31	82.15	97.48
$\boldsymbol{T = 10}$	**89.67**	**82.45**	**97.52**
$T = 20$	89.14	81.82	97.48
$T = 30$	89.08	81.82	97.24

assess the impact of the number of layers T on performance, we trained the SAM [21] added varying CCS layers on the ISIC dataset. As shown in Tab. 3, the optimal T is 10, which we applied across all prior experiments. The CCS layer, derived from the dual algorithm of a variational problem, effectively performs one iteration per layer. Theoretically, the value of T directly influences algorithm convergence. In fact, for $T \leq 10$, the quality of the segmentation results could be improved. While $T > 10$, segmentation result changes are minimal and the model complexity increases, potentially destabilizing backpropagation. Based on this, we recommend setting T within the range of 5–15.

CCS Module as Post-Processing, as Fixed Parameter Layers, or with Learnable Parameters: Our proposed CCS module can be directly used as a post-processing method for segmentation results. Additionally, in our CCS module, the gradient and divergence calculations are implemented using fixed, simple first-order difference convolution kernels, which can alternatively be made learnable. We conduct ablation experiments on these three different configurations.

For the first type, we use the CCS module as a post-processing component for the fine-tuned SAM2 [28], with the prior field provided by the ground-truth field. During the experiments, we found that a higher number of layers T is required to significantly improve the segmentation results. In the post-processing experiments, we set $T = 100$, while in the other two settings, $T = 10$.

Table 4. Ablation Study on different Configurations of CCS module

Dataset	SAM2 [28] with CCS module			*DICE*↑	*IOU*↑	*ACC*↑
	Post	Fixed	Learnable			
Refuge	✓			79.11	66.40	98.14
		✓		81.15	69.39	98.38
			✓	**83.76**	**72.82**	**98.65**
WHU	✓			93.16	88.43	98.58
		✓		93.32	88.56	98.59
			✓	**93.44**	**88.81**	**98.62**

For the third setting, we set the first two layers as fixed, with the following eight convolutional layers being trainable. Results are shown in Tab. 4. In fact, learnable convolutional layers can adaptively adjust their weights during training, enhancing the flexibility of the CCS module and improving the overall segmentation performance.

5. Conclusion

In this work, we introduce the convex combination star (CCS) shape and demonstrate the generality of this shape. To incorporate this shape prior into image segmentation models, we propose the sufficient condition for preserving CCS shape. Additionally, we develop a solution algorithm for the model integrating CCS prior. Since the solution of the Algorithm 1 takes the form of a sigmoid function, we unroll the algorithm into network modules, replacing the conventional sigmoid activation function in neural networks. Thus, the shape information can be effectively integrated with image features via the CCS module. In this paper, we take SAM and SAM2 as an example to show that the proposed methods are very effective. Please refer to the appendix for experimental results on other convolutional networks.

Our approach could be further improved. Theoretically, whether the field conditions can still maintain the CCS when ε does not tend to 0 remains unproven mathematically. In practice, the accurate prediction of the CCS field can be achieved by better designing the predictive network template. We believe that this common shape prior is highly beneficial for precise segmentation in medical, remote sensing, and natural images. It can guide the network to quickly learn the properties of objects.

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